

Operations Research and Information System Management

中英双语课程总结与复习提纲

Based on OR Class 1-13 slides

2026-05-31

目录

1 课程总览 / Course Map	4
2 运营管理基础 / Operations Management Basics	5
2.1 输入-转化-输出 / Input-Transformation-Output	5
2.2 设计决策与运行决策 / Design vs. Operation Decisions	5
2.3 模型与权衡 / Models and Trade-offs	5
3 竞争力、战略与生产率 / Competitiveness, Strategy, and Productivity	6
3.1 竞争力与战略 / Competitiveness and Strategy	6
3.2 生产率 / Productivity	6
4 预测 / Forecasting	7
4.1 好预测的特征 / Features of Good Forecasts	7
4.2 预测过程 / Forecasting Process	7
4.3 定性预测 / Qualitative Forecasting	7
4.4 时间序列方法 / Time-Series Methods	8
4.5 线性趋势与回归 / Linear Trend and Regression	8
4.6 预测误差 / Forecast Accuracy	8
5 产品与服务设计 / Product and Service Design	9
5.1 设计目标 / Design Objectives	9
5.2 标准化、延迟差异化与模块化 / Standardization, Delayed Differentiation, and Mod- ularity	9

5.3	可靠性与稳健设计 / Reliability and Robust Design	10
5.4	产品开发流程 / Product Development Process	10
5.5	并行工程与 DFM / Concurrent Engineering and DFM	10
5.6	设计评审工具 / Design Review Tools	11
5.7	服务设计 / Service Design	11
6	决策理论 / Decision Theory	11
6.1	决策要素 / Decision Elements	11
6.2	决策环境 / Decision Environments	12
6.3	不确定性决策准则 / Criteria Under Uncertainty	12
6.4	风险下的期望值 / Expected Monetary Value	12
6.5	决策树与完全信息价值 / Decision Trees and EVPI	12
6.6	敏感性分析 / Sensitivity Analysis	13
7	产能规划 / Capacity Planning	13
7.1	产能概念 / Capacity Concepts	13
7.2	产能战略 / Capacity Strategies	13
7.3	产能规划步骤 / Capacity Planning Steps	14
7.4	加工需求计算 / Processing Requirements	14
7.5	服务产能与瓶颈 / Service Capacity and Bottlenecks	14
7.6	约束管理 / Constraint Management	15
7.7	成本-产量分析 / Cost-Volume Analysis	15
8	流程选择与设施布局 / Process Selection and Facility Layout	15
8.1	流程类型 / Process Types	15
8.2	自动化 / Automation	16
8.3	基础布局类型 / Basic Layout Types	16
8.4	生产线平衡 / Line Balancing	16
8.5	流程布局成本 / Process Layout Transportation Cost	17
9	高频计算模板 / Calculation Templates	17
9.1	预测题 / Forecasting Problem	17
9.2	决策表 / Decision Table	17
9.3	产能与盈亏平衡 / Capacity and BEP	18

9.4 生产线平衡 / Line Balancing	18
10 复习优先级 / Review Priorities	18
11 中英关键词表 / Bilingual Glossary	18

1 课程总览 / Course Map

Class	中文重点	English Focus
1	课程介绍、评分、项目要求、参考书。	Course orientation, grading, group project, references.
2	运营管理基本概念：输入、转化、输出；系统设计与系统运行；模型和权衡。	OM fundamentals: inputs, transformation, outputs; system design and operation decisions; models and trade-offs.
3	竞争力、战略、订单资格要素/赢单要素、生产率。	Competitiveness, strategy, order qualifiers/winners, productivity.
4	预测（一）：定性预测、时间序列、移动平均、指数平滑、趋势和回归。	Forecasting I: qualitative methods, time series, moving average, exponential smoothing, trend and regression.
5	预测（二）：MAD、MSE、MAPE、控制图、跟踪信号、Excel 方法选择。	Forecasting II: MAD, MSE, MAPE, control charts, tracking signal, Excel-based technique selection.
6	产品与服务设计（一）：标准化、延迟差异化、模块化、可靠性、稳健设计。	Product and service design I: standardization, delayed differentiation, modular design, reliability, robust design.
7	产品与服务设计（二）：开发流程、并行工程、DFM、再制造、FMEA/FTA/VA/QFD/Kano。	Product and service design II: development process, concurrent engineering, DFM, remanufacturing, FMEA/FTA/VA/QFD/Kano.
8	产品与服务设计（三）：服务设计、服务蓝图、高/低接触服务。	Product and service design III: service design, service blueprinting, high-contact vs. low-contact services.
9	决策理论：不确定性、风险、EMV、决策树、EVPI、敏感性分析。	Decision theory: uncertainty, risk, EMV, decision trees, EVPI, sensitivity analysis.
10	产能规划（一）：产能度量、效率、利用率、产能缓冲、产能需求计算。	Capacity planning I: capacity measures, efficiency, utilization, capacity cushion, processing requirements.

11	产能规划 (二): 服务产能、外包、瓶颈、规模经济、约束管理、成本-产量分析。	Capacity planning II: service capacity, outsourcing, bottlenecks, economies of scale, constraints, cost-volume analysis.
12	流程选择与设施布局 (一): 流程类型、产品-流程矩阵、自动化、基础布局类型。	Process selection and layout I: process types, product-process matrix, automation, basic layout types.
13	流程选择与设施布局 (二): 单元布局、生产线平衡、流程布局运输成本。	Process selection and layout II: cellular layout, line balancing, process-layout transportation cost.

2 运营管理基础 / Operations Management Basics

运营管理 (Operations Management) 是把输入转化为产品、服务或信息输出的管理活动。

English: Operations management manages the transformation of inputs into outputs that create value.

2.1 输入-转化-输出 / Input-Transformation-Output

- **中文:** 输入包括劳动力、设备、设施、原材料、时间, 也包括知识、技能、客户关系和声誉。
- **English:** Inputs include labor, equipment, facilities, raw materials, time, knowledge, skills, customer relationships, and reputation.
- **中文:** 输出包括产品、服务和信息。运营的目标是通过转化过程创造价值。
- **English:** Outputs include products, services, and information. The transformation process should add value.

2.2 设计决策与运行决策 / Design vs. Operation Decisions

系统设计决策 / System Design	系统运行决策 / System Operation
产能、选址、设施布局、产品/服务设计、设备购买与布置。	人员管理、库存控制、排程、项目管理、质量保证。
Strategic, long-term commitments that define system parameters.	Tactical and operational decisions that run the system day to day.

2.3 模型与权衡 / Models and Trade-offs

- **模型** (Model): 对现实的抽象和简化, 用于组织信息、量化问题和进行 “what-if” 分析。

- **权衡** (Trade-off): 以放弃某一目标的一部分来换取另一目标, 例如成本、质量、速度、柔性、库存之间的平衡。
- **中文**: 模型不能保证最优决策; 关键是理解用途、假设、限制和结果解释。
- **English**: A model does not guarantee a good decision; managers must understand its purpose, assumptions, limitations, and interpretation.

3 竞争力、战略与生产率 / Competitiveness, Strategy, and Productivity

3.1 竞争力与战略 / Competitiveness and Strategy

竞争力 (Competitiveness) 指企业相对于竞争者满足顾客需求的有效程度。

运营战略 (Operations Strategy) 是与组织战略一致、用于指导运营职能的方案。

- **订单资格要素** (Order Qualifiers): 顾客认为进入备选集合所需的最低标准。
- **赢单要素** (Order Winners): 使产品或服务被顾客认为优于竞争者的特征。
- **中文**: 战略形成通常包括: 定义主要任务、评估独特能力、识别订单资格/赢单要素、确定企业定位。
- **English**: Strategy formulation defines the primary task, assesses distinctive competencies, identifies qualifiers/winners, and positions the firm.

3.2 生产率 / Productivity

$$\text{Productivity} = \frac{\text{Outputs}}{\text{Inputs}}$$

$$\text{Productivity Growth} = \frac{\text{Current Productivity} - \text{Previous Productivity}}{\text{Previous Productivity}}$$

- **中文**: 单要素生产率只看一个输入, 如每劳动小时产出。
- **English**: Partial productivity uses one input, such as output per labor hour.
- **中文**: 多要素生产率使用多个输入, 如劳动力 + 机器 + 资本。
- **English**: Multifactor productivity uses multiple inputs, such as labor, machine, and capital.
- **中文**: 总生产率使用全部投入。
- **English**: Total productivity compares goods/services produced with all inputs used.

4 预测 / Forecasting

4.1 好预测的特征 / Features of Good Forecasts

- **中文:** 及时、准确、可靠、以有意义的单位表达、书面化、容易理解、成本有效。
- **English:** Good forecasts are timely, accurate, reliable, meaningful, written, understandable, and cost-effective.
- **中文:** 预测通常假设过去的因果系统在未来仍有一定延续性。
- **English:** Forecasting often assumes that the causal system in the past continues into the future.

4.2 预测过程 / Forecasting Process

1. 确定预测目的 / Determine the purpose.
2. 建立时间范围 / Establish the time horizon.
3. 选择预测技术 / Select a forecasting technique.
4. 收集并分析数据 / Gather and analyze data.
5. 做出预测 / Make the forecast.
6. 监控预测 / Monitor the forecast.

4.3 定性预测 / Qualitative Forecasting

方法 / Method	中文说明	English Note
Executive opinions	高层管理者小组判断，快但可能受偏见影响。	Senior managers' judgment; fast but can be biased.
Sales force opinions	销售人员接近客户，信息直接，但可能有激励偏差。	Salespeople know customers directly, but incentives can bias estimates.
Consumer survey	问卷收集顾客意向，适合新产品，但成本高。	Surveys collect customer intentions, useful for new products but costly.
Delphi method	多轮专家问卷，匿名反馈，减少从众压力。	Iterative expert questionnaires with anonymous feedback to reduce group pressure.
Outside opinions	第三方或外部机构预测，省力但依赖外部质量。	External expert or third-party forecasts; convenient but dependent on provider quality.

4.4 时间序列方法 / Time-Series Methods

$$\text{Naive: } F_t = A_{t-1}$$

$$\text{Moving Average: } F_t = \frac{A_{t-1} + A_{t-2} + \cdots + A_{t-n}}{n}$$

$$\text{Weighted Moving Average: } F_t = \sum_{i=1}^n w_i A_{t-i}, \quad \sum_{i=1}^n w_i = 1$$

$$\text{Exponential Smoothing: } F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})$$

- **中文:** α 越小, 预测越平滑但滞后更明显; α 越接近 1, 预测越快跟随实际值。
- **English:** A smaller α creates smoother forecasts with more lag; a larger α responds more quickly to actual demand.
- **中文:** 当序列有明显线性趋势时, 用趋势投影或回归比简单平滑更合适。
- **English:** When a clear linear trend exists, trend projection or regression is more appropriate than simple smoothing.

4.5 线性趋势与回归 / Linear Trend and Regression

$$F_t = a + bt$$

$$b = \frac{n \sum ty - \sum t \sum y}{n \sum t^2 - (\sum t)^2}, \quad a = \frac{\sum y - b \sum t}{n}$$

最小二乘法 (Least Squares) 选择使误差平方和最小的直线:

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

4.6 预测误差 / Forecast Accuracy

$$e_t = A_t - F_t$$

$$MAD = \frac{\sum |A_t - F_t|}{n}$$

$$MSE = \frac{\sum (A_t - F_t)^2}{n - 1}$$

$$MAPE = \frac{100}{n} \sum \left| \frac{A_t - F_t}{A_t} \right|$$

$$\text{Tracking Signal} = \frac{\sum (A_t - F_t)}{MAD}$$

- **中文:** MAD 线性惩罚误差, 容易解释。
- **English:** MAD weights errors linearly and is easy to interpret.

- **中文:** MSE 对大误差惩罚更重, 适合强调大偏差风险。
- **English:** MSE penalizes large errors more strongly.
- **中文:** MAPE 用百分比表达, 便于跨规模比较。
- **English:** MAPE expresses error as a percentage and supports comparison across scales.
- **中文:** 跟踪信号绝对值过大说明预测可能存在系统性偏差。
- **English:** A large absolute tracking signal indicates possible forecast bias.

5 产品与服务设计 / Product and Service Design

5.1 设计目标 / Design Objectives

- **中文:** 把顾客需求转化为产品/服务要求, 并兼顾成本、质量、可制造性、可维护性、法律伦理和环境。
- **English:** Translate customer needs into product/service requirements while balancing cost, quality, manufacturability, maintainability, legal, ethical, and environmental issues.
- **中文:** 设计既要面向市场, 也要考虑运营能力。
- **English:** Design must fit both market requirements and operational capabilities.

5.2 标准化、延迟差异化与模块化 / Standardization, Delayed Differentiation, and Modularity

概念 / Concept	中文	English
Standardization	减少产品、服务或流程中的变化, 带来规模效率和质量一致性。	Reduces variety in products, services, or processes, improving efficiency and consistency.
Component commonality	在不同产品中使用相同零件, 降低库存、采购和培训复杂度。	Uses common components across products to reduce inventory, purchasing, and training complexity.
Delayed differentiation	先生产但不完全完成, 等顾客偏好明确后再定制。	Produces but postpones final completion until customer preferences are known.

Modular design	把组件划分为可替换模块，便于组装、维修和定制。	Subdivides components into interchangeable modules for assembly, repair, and customization.
Mass customization	以标准化为基础提供一定程度的个性化。	Provides customization while retaining standardized operations.

5.3 可靠性与稳健设计 / Reliability and Robust Design

- **可靠性** (Reliability): 产品、零件或系统在规定条件下完成预期功能的能力。
- **失效** (Failure): 产品、零件或系统没有按预期工作。
- **稳健设计** (Robust Design): 产品或服务能在广泛条件下保持一致性能。
- **中文**: 提升可靠性的方式包括改进组件设计、生产/装配技术、测试、冗余、预防性维护和用户教育。
- **English**: Reliability can be improved through component design, production/assembly techniques, testing, redundancy, preventive maintenance, and user education.

5.4 产品开发流程 / Product Development Process

1. Idea generation / 创意产生
2. Feasibility analysis / 可行性分析
3. Product specifications / 产品规格
4. Process specifications / 流程规格
5. Prototype development / 原型开发
6. Design review / 设计评审
7. Market test / 市场测试
8. Product introduction / 产品导入

5.5 并行工程与 DFM / Concurrent Engineering and DFM

并行工程 (Concurrent Engineering) 让设计、运营、制造、营销等人员早期共同参与，减少“over-the-wall”顺序开发造成的返工。

可制造性设计 (Design for Manufacturing, DFM) 强调产品设计要便于经济制造、装配和质量控制。

5.6 设计评审工具 / Design Review Tools

工具 / Tool	中文用途	English Use
FMEA	系统分析失效模式、原因、影响和纠正措施。	Analyzes failure modes, causes, effects, and corrective actions.
FTA	用树形图分析故障之间的逻辑关系。	Uses a tree diagram to analyze relationships among failures.
VA	识别并去除不必要功能或过高成本。	Eliminates unnecessary features and excessive cost.
QFD / House of Quality	把顾客之声转化为设计特性、目标值和竞争评估。	Translates the voice of the customer into design characteristics, targets, and competitive assessment.
Kano Model	区分基本需求、绩效需求和兴奋需求。	Classifies basic, performance, and excitement requirements.

5.7 服务设计 / Service Design

- **服务概念** (Service Concept): 服务目的、目标市场和顾客体验。
- **服务包** (Service Package): 实物项目、感官利益和心理利益的组合。
- **服务规格** (Service Specifications): 从绩效规格转化为设计规格和交付规格。
- **服务蓝图** (Service Blueprinting): 描述和分析服务交付系统的流程工具。

服务蓝图步骤 / Major Service Blueprinting Steps

1. Establish boundaries / 确定边界
2. Identify sequence of customer interactions / 识别顾客互动顺序
3. Develop time estimates / 估计时间
4. Identify potential failure points / 识别潜在失效点
5. Establish a time frame / 建立时间框架
6. Analyze profitability / 分析盈利性

6 决策理论 / Decision Theory

6.1 决策要素 / Decision Elements

- **中文**: 自然状态: 未来可能发生且影响收益的外部条件。

- **English:** States of nature are future conditions that affect payoffs.
- **中文:** 备选方案：管理者可以选择的行动。
- **English:** Alternatives are the actions available to the decision maker.
- **中文:** 收益表：每个方案在每个自然状态下的结果。
- **English:** A payoff table records the result of each alternative under each state of nature.

6.2 决策环境 / Decision Environments

环境 / Environment	中文	English
Certainty	相关参数确定，结果已知。	Relevant parameters and outcomes are known.
Uncertainty	自然状态概率未知。	Probabilities of states of nature are unknown.
Risk	自然状态概率已知或可估计。	Probabilities are known or can be estimated.

6.3 不确定性决策准则 / Criteria Under Uncertainty

- **Maximin** (:) 选择最坏结果中最好的方案；保守。
- **Maximax** (:) 选择最好结果中最好的方案；乐观。
- **Laplace** (:) 假设各自然状态等可能，选择平均收益最高的方案。
- **Minimax Regret** (:) 先构造后悔值表，再选择最大后悔值最小的方案。

6.4 风险下的期望值 / Expected Monetary Value

$$EMV_i = \sum_s p_s \times \text{Payoff}_{i,s}$$

中文: 选择 EMV 最大的方案。

English: Choose the alternative with the highest EMV.

6.5 决策树与完全信息价值 / Decision Trees and EVPI

- **中文:** 决策树从左往右读，从右往左分析。
- **English:** Read a decision tree from left to right; analyze it from right to left.
- **中文:** 完全信息期望价值衡量若能提前知道自然状态，最多愿意为信息支付多少。

- **English:** Expected Value of Perfect Information measures the maximum value of knowing the state of nature before deciding.

$$EVPI = A - B$$

$$A = \sum_s p_s \max_i (\text{Payoff}_{i,s}), \quad B = \max_i EMV_i$$

6.6 敏感性分析 / Sensitivity Analysis

中文: 敏感性分析确定某一概率或参数在什么范围内变化时, 当前最优方案仍保持最优。

English: Sensitivity analysis determines the parameter or probability range over which the current best alternative remains best.

7 产能规划 / Capacity Planning

7.1 产能概念 / Capacity Concepts

概念 / Concept	中文	English
Design capacity	系统设计允许的最大产出率或服务能力。	Maximum designed output rate or service capacity.
Effective capacity	设计产能扣除维护、休息、废品等后的能力。	Design capacity minus allowances such as maintenance, personal time, and scrap.
Actual output	实际实现的产出率, 不能超过有效产能。	Output actually achieved; cannot exceed effective capacity.
Capacity cushion	为需求不确定性预留的额外产能。	Extra capacity used to offset demand uncertainty.

$$\text{Efficiency} = \frac{\text{Actual Output}}{\text{Effective Capacity}}$$

$$\text{Utilization} = \frac{\text{Actual Output}}{\text{Design Capacity}}$$

$$\text{Capacity Cushion} = 100\% - \text{Utilization}$$

7.2 产能战略 / Capacity Strategies

- **Leading / Expansionist** (:): 在需求增长前提前增加产能, 降低缺货风险但可能产能闲置。

- **Following / Wait-and-see** (:) 等需求超过现有产能后再扩张, 降低闲置但可能错失需求。
- **Tracking** (:) 小幅、频繁增加产能, 尽量贴近需求增长。

7.3 产能规划步骤 / Capacity Planning Steps

1. Estimate future capacity requirements / 估计未来产能需求
2. Evaluate existing capacity / 评估现有产能
3. Identify alternatives / 识别备选方案
4. Conduct financial analysis / 财务分析
5. Assess qualitative issues / 评估定性因素
6. Select one alternative / 选择方案
7. Implement / 实施
8. Monitor results / 监控结果

7.4 加工需求计算 / Processing Requirements

$$NR = \frac{\sum_{i=1}^k p_i D_i}{T}$$

- p_i : 每单位产品 i 所需标准加工时间 / standard processing time per unit of product i .
- D_i : 产品 i 的需求量 / demand for product i .
- T : 单台设备或单个资源可用时间 / available time per resource.
- **中文**: 若题目给产能缓冲, 需要把需求或需求产能按 $1 + \text{cushion}$ 调整。
- **English**: If a capacity cushion is required, adjust required capacity by $1 + \text{cushion}$.

7.5 服务产能与瓶颈 / Service Capacity and Bottlenecks

- **中文**: 服务产能通常必须靠近顾客, 且服务不能储存, 因此时间匹配比制造更关键。
- **English**: Service capacity often must be near customers and cannot be stored, so timing is critical.
- **瓶颈** (Bottleneck): 限制系统总产出的工序或资源。
- **中文**: 增加瓶颈之前的产能通常不能提高系统产出; 应优先提升瓶颈本身或重排流程。
- **English**: Adding capacity before a bottleneck usually does not improve total output; improve the bottleneck itself or rebalance the process.

7.6 约束管理 / Constraint Management

1. Identify the most pressing constraint / 识别最紧约束
2. Change the operation to achieve maximum benefit / 调整运作以最大化约束资源产出
3. Make sure other portions support the constraint / 让其他环节支持约束
4. Explore and evaluate alternatives / 探索替代方案
5. Repeat after one constraint is resolved / 一个约束解决后继续循环

7.7 成本-产量分析 / Cost-Volume Analysis

$$TC = FC + vc \times Q$$

$$TR = R \times Q$$

$$P = TR - TC = Q(R - vc) - FC$$

$$Q_{BEP} = \frac{FC}{R - vc}$$

$$Q_{\text{target profit}} = \frac{P_{\text{target}} + FC}{R - vc}$$

中文：多方案比较时，分别写出各方案总成本函数，求交点并结合可行产量区间判断。

English: For multiple alternatives, write each total-cost function, solve intersections, and check feasible volume ranges.

8 流程选择与设施布局 / Process Selection and Facility Layout

8.1 流程类型 / Process Types

类型 / Type	中文	English
Job shop	小批量、高变化、高柔性。	Low volume, high variety, high flexibility.
Batch	中等产量和中等变化。	Moderate volume and moderate variety.
Repetitive / assembly line	高产量、标准化产品或服务。	High volume of standardized goods or services.
Continuous flow	极高产量、非离散产品。	Very high volume of non-discrete goods.
Project	非常规、一次性、位置固定或复杂任务。	Non-routine, one-time, often fixed-position or complex work.

8.2 自动化 / Automation

- **固定自动化** (Fixed Automation): 高产量、低变化, 变更成本高。
- **程序自动化** (Programmable Automation): 计算机程序控制, 适合小批量多品种。
- **柔性自动化** (Flexible Automation): 更广品种和中高产量, 结合 CNC、CAM、CIM 等。
- **中文:** 自动化优点是低变异、低单位变动成本; 缺点是固定成本高、改变困难、可能影响士气。
- **English:** Automation reduces variability and variable cost, but has high fixed cost, lower flexibility, and possible morale effects.

8.3 基础布局类型 / Basic Layout Types

布局 / Layout	适用情境	Key Trade-off
Product layout	标准化加工、高产量、连续或重复流程。	High output and low unit cost, but inflexible and vulnerable to shutdowns.
Process layout	多样化加工需求、作业车间或批量流程。	Flexible and robust to equipment failure, but routing and material handling are complex.
Fixed-position layout	产品或项目太大、太重或不可移动, 如基础设施。	Product stays stationary; workers, materials, and equipment move.
Cellular layout	把机器按相似加工需求组成单元, 兼顾柔性和效率。	Groups part families to reduce movement and improve flow.

8.4 生产线平衡 / Line Balancing

生产线平衡 (Line Balancing) 是把任务分配到工作站, 使各站时间要求尽量相等, 以提高劳动力和设备利用率。

$$\text{Output Rate} = \frac{OT}{CT}$$

$$CT = \frac{OT}{D}$$

$$N_{\min} = \left\lceil \frac{\sum t_i}{CT} \right\rceil$$

$$\text{Percent Idle Time} = \frac{\text{Idle time per cycle}}{N \times CT}$$

$$\text{Efficiency} = 1 - \text{Percent Idle Time}$$

- *OT*: 每日可作业时间 / operating time per day.
- *CT*: 周期时间 / cycle time.
- *D*: 目标产出 / desired output.
- *N*: 工作站数量 / number of workstations.
- **中文**: 分配任务时必须满足前置关系, 并且任务时间不能超过当前工作站剩余时间。
- **English**: Task assignment must respect precedence and fit within the remaining station time.

8.5 流程布局成本 / Process Layout Transportation Cost

$$\text{Total Load-Distance Cost} = \sum_{i < j} F_{ij} D_{ij} C_{ij}$$

- F_{ij} : 部门 i, j 的物流量 / workflow between departments.
- D_{ij} : 分配位置之间的距离 / distance between assigned locations.
- C_{ij} : 单位运输成本 / unit movement cost. 若题目未给, 可默认为 1。
- **中文**: 目标是把物流量大的部门放得更近, 最小化运输时间、距离或成本。
- **English**: The goal is to place high-flow departments closer together and minimize travel time, distance, or cost.

9 高频计算模板 / Calculation Templates

9.1 预测题 / Forecasting Problem

1. **中文**: 写出实际值 A_t 、预测值 F_t 和误差 $e_t = A_t - F_t$ 。
2. **中文**: 根据题目选择 naive、MA、WMA、exponential smoothing 或 trend。
3. **中文**: 计算 MAD/MSE/MAPE 并比较方法。
4. **English**: Write actuals, forecasts, and errors; apply the requested method.
5. **English**: Compute MAD/MSE/MAPE and compare alternatives.

9.2 决策表 / Decision Table

1. **中文**: 列出方案、自然状态和收益表。
2. **中文**: 不确定性: 按 Maximin、Maximax、Laplace 或 Minimax Regret。
3. **中文**: 风险: 计算每个方案 EMV 。
4. **中文**: $EVPI$: 先算完全信息下的期望值 A , 再减去最佳 EMV 。
5. **English**: List alternatives, states, and payoffs; choose the criterion; compute EMV or $EVPI$ as needed.

9.3 产能与盈亏平衡 / Capacity and BEP

1. **中文:** 识别 design capacity、effective capacity、actual output。
2. **中文:** 计算 efficiency、utilization、capacity cushion。
3. **中文:** 写出 TC 、 TR 、profit, 求 Q_{BEP} 或目标利润产量。
4. **English:** Identify capacity measures, compute efficiency/utilization/cushion, and solve cost-volume equations.

9.4 生产线平衡 / Line Balancing

1. **中文:** 根据目标产出求 $CT = OT/D$ 。
2. **中文:** 根据总任务时间求 $N_{\min}$ 。
3. **中文:** 画前置图, 按启发式规则分配任务。
4. **中文:** 计算 idle time 和 efficiency。
5. **English:** Compute cycle time, minimum workstations, assign tasks by precedence and heuristic rules, then compute idle time and efficiency.

10 复习优先级 / Review Priorities

- Forecasting formulas / 预测公式 利用率
- MAD, MSE, MAPE, tracking signal / 误差指标 • BEP and make-or-buy / 盈亏平衡与自制外购
- Decision criteria / 决策准则 • Process types and layouts / 流程类型和布局
- EMV and EVPI / 期望值与完全信息价值 • Line balancing / 生产线平衡
- Capacity efficiency/utilization / 产能效率与 • QFD, FMEA, Kano / 设计质量工具
- • Service blueprinting / 服务蓝图

11 中英关键词表 / Bilingual Glossary

English	中文	记忆点 / Note
Operations management	运营管理	输入转化为输出并创造价值。
Transformation process	转化过程	Physical, locational, exchange, storage, physiological, informational transformations.
Order qualifier	订单资格要素	进入顾客备选集合的最低标准。
Order winner	赢单要素	真正促成选择的竞争优势。

Productivity	生产率	Output divided by input.
Forecast error	预测误差	Actual minus forecast.
Exponential smoothing	指数平滑	Previous forecast plus alpha times error.
Tracking signal	跟踪信号	Detects persistent bias.
Standardization	标准化	Efficiency vs. reduced variety.
Delayed differentiation	延迟差异化	Postpone final customization.
Modular design	模块化设计	Interchangeable modules.
Reliability	可靠性	Performs intended function under specified conditions.
Robust design	稳健设计	Performs across broad operating conditions.
FMEA	失效模式与影响分析	Failure mode, cause, effect, action.
QFD	质量功能展开	Voice of customer to design requirements.
Kano model	Kano 模型	Basic, performance, excitement needs.
State of nature	自然状态	Uncontrollable future condition.
Maximin	最大最小准则	Conservative best-worst choice.
Minimax regret	最小最大后悔准则	Minimize worst regret.
EMV	期望货币价值	Weighted average payoff.
EVPI	完全信息期望价值	Maximum value of perfect information.
Design capacity	设计产能	Maximum designed output rate.
Effective capacity	有效产能	Design capacity minus allowances.
Capacity cushion	产能缓冲	100 percent minus utilization.
Bottleneck	瓶颈	Constraint on system output.
Break-even point	盈亏平衡点	Revenue equals total cost.
Job shop	作业车间	Low volume, high variety.
Batch	批量流程	Moderate volume and variety.
Product layout	产品布局	Sequential standardized flow.

Process layout	流程布局	Functional grouping.
Cellular layout	单元布局	Group technology by part family.
Line balancing	生产线平衡	Assign tasks to stations evenly.
